Labs - Innovating Product Development with 3D Printing and Additive Manufacturing

TGS-2022017597 | one folder per lab | full step-by-step in each README and in the Learner Guide.

Topic 01 - Overview of Additive Manufacturing and 3D Scanning

- Lab 1 - Extrude - Personalised Keychain Token (token.stl)
- Lab 2 - Sweep - Giant Paper Clip Bookmark
- Lab 3 - Loft - Mini Plant Pot with Custom Rails
- Lab 4 - Revolve - Funnel from a 2D Profile
- Lab 5 - Tolerance - Container and Lid (0.2 mm Fit) (FDM-Box-Final.f3d, FDM-Box-Final.stl, FDM-Box-Cura.3mf)
- Lab 6 - 3D Scan and Print a Real Object

Topic 02 - Process Related Issues for AM

- Lab 7 - Install Cura and Slice Your First Model (cube10.stl, cube20.stl)
- Lab 8 - Bed Adhesion - Skirt vs Brim vs Raft (cube10.stl)
- Lab 9 - Support Structures for Overhangs (twist.stl, Twisted-Solid.stl, wedge.stl, cube20-sideways-hole.stl)
- Lab 10 - Filament Diagnostics - Diameter, Temperature and Flow (retraction-test.stl, cooling-spire.stl, twist-with-cooling-tower.stl)

Topic 03 - Process Improvement for AM

- Lab 11 - Perimeters and Infill - Strength vs Speed (cube10.stl, line-width-test.stl, cube20-hole-0-wall.stl, cube20-hole-1-wall.stl)
- Lab 12 - Infill Patterns - Zigzag vs Grid Strength Test (cube10.stl, optimize-1.stl, optimize-2.stl, optimize-3.stl, optimize-4.stl)
- Lab 13 - Layer Height and Top Thickness - Surface Quality (cube10.stl, sloped-cube.stl)
- Lab 14 - Ironing - Glass-Smooth Top Surfaces (cube10.stl, z-block.stl)
- Lab 15 - Fuzzy Skin - Textured Walls on a Vase (open vase.stl, 1.5mm-wall-vase.stl)
- Lab 16 - Post-Processing - Support Removal and Sanding (open vase.stl, test-part.stl)